Ysbyty Gwynedd
Anaesthetics Department
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PROVISION IN REMOTE SITES

Provision of Anaesthesia in Radiology and MRI

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Key Points

- This is a friendly guide to help you familiarise yourself with the area. If anything is not clear, please ask.

MRI Scan Room

If you are asked to work in MRI scan, there is no provision for general anaesthetic or intravenous sedation (no monitoring or anaesthetic equipment that is compatible with MRI). There is an oral sedation protocol for children[a] and adults are advised to use oral benzodiazepines, via the GP, if necessary

CT Scan Room & interventional Suite (IS)

These are the two areas in the Xray department where general anaesthetics may be administered.

General information

Most of the patients coming through the unit are already in-patients or require the procedure sooner rather than later. Many have not benefited from a pre-operative assessment clinic.

The Anaesthetic machine and ABC trolley are stored in the Interventional suite.

There are 3 cardiac arrest trolleys in Radiology- one outside Interventional Suite 0900-1700 Mon to Fri (moved to ED at close of day) and one each permanently in the Recovery areas for both CT and Interventional suite

There is a drug cupboard in the Recovery area of IS which has the most commonly used drugs[b] inside. The Radiology nurses hold the key for this cupboard. Refrigerated medications eg muscle relaxants are held in the fridge in the store room behind IS.

There is a controlled drug cupboard containing fentanyl and midazolam in both CT and IS- access is via nursing staff in X ray (bleep 036)

There is a supply of forced air warmers, though if you require fluid warming, this is available from main theatre

If you need to access IS, then ring 5552 or bleep the radiology nurse on 036

If you need to access the CT scan room, it is manned by Radiographers. Either knock on the control room door or bleep the on call radiographer for CT via switchboard

Interventional suite

The interventional suite is the most commonly used area for general anaesthesia and sedation. A variety of cases are done here, most frequently, to relieve urological, biliary and gastrointestinal obstruction.

There is a Dräger Tiro machine and airway/ breathing/ circulation trolley stored there.

CT scan room

There are two situations in which patients are brought to CT scan room:-

i) Emergency requiring intubation and definitive scanning at any time

These patients must be treated as transfers where the kit necessary is brought with them.

If a patient is believed to be suitable for sedation and it does not work out, take the patient back to a designed area with appropriate equipment

ii) Elective requiring scan in patients who cannot tolerate it (learning disability or children) or patients requiring bone biopsies

These patients are anaesthetised in CT scan using the equipment that is held in IS. There are only about 2 procedures per annum, If anyone ends up doing any cases, then one needs to check that all equipment is available and that you are familiar with room layout and position proposed for patient before starting.

Recovery

There are small recovery areas with standard monitoring, O_2 and suction next to both IR suite and CT scan room. The patients are recovered with the aid of the radiology nurses. The radiology nurses spend a small number of days each year in Main Recovery and are not fully trained in airway management. I expect the anaesthetist to stay with the patient until they are awake, with no airway adjuncts and spontaneously breathing. If there is a clinically challenging case, you may approach Main Recovery for help with the patient.

Ionising Radiation

The radiographers are in charge of the IRMER regulations (ionising radiation) and you will be offered protective clothing (gown, thyroid shield, X ray dense shield) as well as an indicator badge. It is useful for you to know that the penetration of X radiation reduces a great deal the further you are from the source. It is vanishingly small at 2 metres from source

References

AAGBI Guideline MRI - Provision Of Anaesthetic Services In Magnetic Resonance Units published May 2002.

AAGBI MRI update - Safety in Magnetic Resonance Units
published July 2010.